

**AI Model Trading Market for Industry 5.0:
Research on Architecture Design and Incentive
Mechanism**

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Architecture Design and Incentive Mechanism Research

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summary

Industry 5.0, with its core values of human-centeredness, sustainability, and resilience, aims to achieve deep integration and collaboration between humans and machines. In this paradigm shift, AI models trained on edge devices have become core digital assets for enterprises. Building a secure, fair, and efficient model sharing and trading system has become a key challenge for both academic research and industrial practice.

Existing AI model platforms (such as HuggingFace and ModelScope) primarily focus on the public availability and download of open-source models, exhibiting significant shortcomings in areas such as commercial transaction mechanisms, industrial-grade quality verification, intellectual property protection, and fair incentives for participants. To address these issues, this study systematically proposes an AI model trading market architecture and its incentive mechanism for the Industry 5.0 environment, and conducts comprehensive verification through prototype system implementation and multi-dimensional experiments.

At the architectural design level, this paper proposes a five-layer hybrid architecture that organically integrates key technologies such as edge computing, federated learning, blockchain, differential privacy, and secure multi-party computation. At the mechanism design level, a three-level quality contract menu is constructed based on contract theory to address the information asymmetry problem, and the Shapley value is introduced to realize multi-dimensional contribution evaluation and fair benefit distribution.

Experimental results show that the proposed method improves the accuracy by up to 3.17 percentage points compared to FedAvg in Non-IID data environments. The incentive mechanism based on the Shapley value reduces the Gini coefficient to 0.285 (a 41.5% reduction compared to the no-incentive scheme). The application of differential privacy ($\epsilon=1.0$) reduces the success rate of member inference attacks from 68.3% to 12.3%, while the system stably maintains a processing throughput of 820 TPS and a transaction success rate of 95.8% in a 100-node environment.

This study provides a complete theoretical framework, architectural design scheme, and experimental verification for the AI model trading market under the distributed collaboration paradigm of Industry 5.0, which has important theoretical value and

practical guiding significance for promoting the construction of a safe, fair, and sustainable industrial AI ecosystem.

Keywords: Industry 5.0; AI model trading market; Federated learning; Blockchain; Incentive mechanism; Differential privacy; Shapley value

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Chapter 1 Introduction

1.1 Research Background and Significance

1.1.1 Development Trends of Industry 5.0 and Human-Machine Collaboration

In 2021, the European Commission released the official report "Industry 5.0: Towards a Sustainable, Human-centric and Resilient European Industry", formally proposing the concept of Industry 5.0 [1]. Industry 5.0 surpasses the production automation and hyper-connectivity pursued by Industry 4.0, and establishes human-centricity, sustainability and resilience as the three core value principles of industrial system design. This is not only an iterative evolution at the technological level, but also a fundamental redefinition of the roles of humans and machines – from machines replacing humans to human-machine collaborative co-creation.

Industry 4.0 focuses on automating and maximizing the efficiency of the production process through cyber-physical systems (CPS), the Internet of Things (IoT) and big data analytics; while Industry 5.0 emphasizes the deep integration of human creativity, critical thinking and situational adaptability with the precision, repeatability and massive data processing capabilities of machines [2]. Under this new paradigm of human-machine collaboration, artificial intelligence systems need to undertake the following core functions: (1) realize real-time reasoning and dynamically adapt to changes in the production environment on edge devices; (2) support the secure sharing and collaborative

evolution of models across enterprises and devices; (3) protect the core data assets and intellectual property rights of enterprises and models; and (4) establish an effective incentive mechanism to encourage all participants to actively contribute high-quality models and data resources.

1.1.2 The Rise of the AI Model Trading Market

With the rapid development of deep learning technology, trained AI models have evolved from simple software components into key digital assets that determine the core competitiveness of enterprises. According to a McKinsey (2023) report, the global AI market size is expected to reach approximately \$190 billion in 2025, with the proportion of transactions and licensing of pre-trained models continuing to expand [3].

At present, a variety of AI model platforms have formed an active model sharing ecosystem. Internationally, HuggingFace hosts more than 300,000 models, and platforms such as Kaggle Models and Civitai are also booming. Domestically, ModelScope, Baidu Qianfan, and iFlytek Open Platform are becoming increasingly mature [4]. However, in-depth analysis reveals the following structural limitations of these platforms: First, they are dominated by open-source and free distribution, lacking sustainable commercial transactions and value realization mechanisms; Second, they are designed for general scenarios and lack vertical solutions for special needs in industrial sites (millisecond-level real-time inference, reliability in extreme environments, and heterogeneity of domain data); Third, the intellectual property protection and anti-counterfeiting traceability mechanisms of models are weak, making it difficult to meet the protection needs of enterprises for core model

assets; Fourth, an effective mechanism has not yet been established to quantitatively evaluate the multidimensional contributions of participants and incentivize long-term high-quality participation.

1.1.3 Necessity of the Research

Based on the above analysis, conducting research on the AI model trading market for the Industry 5.0 environment is necessary and urgent on multiple levels and dimensions:

Theoretical level: Existing research focuses on single technical dimensions such as federated learning algorithm optimization[5][6] or blockchain data transaction mechanism[7][8], and lacks results on the systematic research of integrating these technologies into a complete AI model transaction market architecture. Especially under the complex conditions of information asymmetry and participant heterogeneity, how to construct a theoretical framework that combines incentive compatibility and distributive fairness is still in its early stages of academic exploration.

In practice: Industrial enterprises face strict restrictions from data privacy regulations (such as GDPR and data security laws), making direct data sharing almost impossible. However, they urgently need to leverage collaborative learning and model exchange to improve AI model performance. The demand for edge AI models in manufacturing, logistics, energy management, and other fields is growing rapidly, but the lack of a secure, efficient, and fair model sharing mechanism has become a core bottleneck restricting the promotion of collaborative intelligence in industry.

From a technical perspective: Realizing an AI model trading market requires a deep integration of interdisciplinary technologies—blockchain provides a decentralized trust foundation, federated learning ensures collaborative training with data remaining within its domain, privacy computing (differential privacy, secure multi-party computation) protects the confidentiality of data and models, and game theory provides a rigorous mathematical foundation for designing incentive-compatible mechanisms. The complexity of this technological integration itself constitutes a scientific problem of significant research value.

1.2 Current Status of Research at Home and Abroad

This research sits at the intersection of four active research lines: federated learning and distributed model training, AI model markets and trading platforms, blockchain-driven model verification and traceability, and incentive mechanisms and game theory methods.

1.2.1 Federated Learning and Distributed Model Training

The FedAvg algorithm proposed by McMahan et al. (2017) established the basic paradigm of federated learning. It achieved collaborative training without data leaving the domain by using a weighted average aggregation of local model updates on the client through a central server [5].

Subsequently, FedProx by Li et al. (2020) introduced proximal terms to solve the system heterogeneity problem [9]; SCAFFOLD by Karimirdy et al. (2020) used control variables to alleviate client drift [10]; and FedNova by Wang et al. (2020) improved convergence in the Non-IID environment by normalizing the difference in the number of local update steps of different clients [11]. Kairouz et al. (2021) systematically summarized the open challenges of federated learning in a comprehensive review, in which they clearly pointed out that incentive mechanisms and

model evaluation are key issues that have not yet been fully resolved [6]. The above studies mainly focus on performance optimization at the algorithm level, while insufficient attention is paid to the core issues affecting the sustainability of the federated learning ecosystem, such as the measurement of participants' contributions and economic incentives.

1.2.2 AI Model Market and Trading Platform

Research on assetizing AI models and building trading platforms has become increasingly active in recent years. Chen et al. (2020) proposed FedHealth, a health monitoring framework based on federated transfer learning, and initially explored the technical path of cross-institutional model sharing [12]. Liu et al. (2022) proposed the GTG-Shapley method for evaluating the contribution of participants in federated learning based on the Shapley value of data. It achieved a good balance between computational efficiency and evaluation accuracy, but its research scenario was limited to contribution evaluation in the training stage and failed to extend to the complete model trading market [13]. In terms of commercial platforms, HuggingFace and ModelScope have achieved remarkable results in the model release ecosystem, but as mentioned above, their verticalization and commercialization capabilities for the specific needs of Industry 5.0 are still insufficient.

1.2.3 Blockchain-Driven Model Validation and Incentive Mechanism

The application of blockchain technology in AI model governance mainly revolves around integrity verification and trusted traceability. Zhang et al. (2021) proposed a model version management and audit trail scheme based on Hyperledger Fabric [14]; Kumar et al. (2021) designed a smart

contract-driven automated model authorization and royalty distribution mechanism [15]. In terms of incentive mechanisms, Kang et al. (2019) combined reputation systems with contract theory and proposed a joint optimization incentive framework for reliable federated learning, and verified its effectiveness in improving participation rate and suppressing malicious behavior through simulation [16]. Wang et al. (2021) used contract theory to conduct a rigorous formal modeling of the incentive problem of secure federated learning in the scenario of drone-assisted crowd perception [17]. The above studies provide an important methodological foundation for this work, but a comprehensive study that organically integrates blockchain notarization, contract theory incentives and AI model market is still lacking.

1.2.4 Limitations of existing research and the innovativeness of this study

Based on the above literature analysis, the following common research gaps can be identified: (1) There is a wealth of research on single-point technologies, but there is a lack of systematic work that unifies federated learning, blockchain, privacy computing and game theory into a complete AI model trading market architecture; (2) Most studies are based on general scenarios and fail to fully respond to the special constraints of Industry 5.0 – real-time performance, edge heterogeneity, non-IID data distribution and strict compliance requirements; (3) Model quality evaluation, privacy protection and incentive allocation are often studied in isolation and lack an integrated comprehensive framework.

To address the aforementioned gaps, this research is innovative in three aspects. First, it proposes and designs in detail a five-layer hybrid

architecture integrating edge computing, federated learning, privacy computing, and blockchain technologies, providing a complete systematic reference design for the AI model trading market in an Industry 5.0 environment. Second, it innovatively integrates contract theory and Shapley value into a two-layer dynamic incentive mechanism. The first layer addresses the provider self-selection problem under information asymmetry through a contract menu, while the second layer achieves fair allocation precisely matched with marginal contributions through Shapley value. Third, through prototype system implementation and systematic multi-dimensional experiments, it conducts the first empirical quantitative analysis of the ternary trade-off between model performance, privacy protection, and transaction credibility.

1.3 Research Content and Innovations

The overall goal of this research is to design an AI model trading market architecture adapted to the Industry 5.0 environment, develop an incentive mechanism that motivates participants to make continuous high-quality contributions, and conduct comprehensive empirical verification through a prototype system. The specific research content covers the following six aspects:

- (1) A five-layer hybrid architecture for the AI model trading market that meets the core requirements of Industry 5.0 (real-time performance, security, and scalability) is proposed.
- (2) Based on contract theory, design a three-level quality contract menu and solve the optimal incentive contract under information asymmetry.

- (3) Construct a four-dimensional contribution evaluation model that integrates model quality, data contribution, computing resources and reputation.
- (4) Design and implement a fair revenue distribution mechanism based on the TMC-Shapley approximation algorithm.
- (5) Construct an integrated privacy protection framework based on differential privacy and secure multi-party computation, and analyze the privacy-utility trade-off.
- (6) Develop a prototype system and conduct comprehensive verification through multi-dimensional experiments on federated learning performance, incentive fairness, privacy protection effect and system scalability.

The main innovations can be summarized as follows: (1) The first proposed layered hybrid architecture design paradigm of distributed control + centralized execution was proposed; (2) Contract theory and Shapley value were integrated into a two-layer incentive mechanism to solve the problems of information asymmetry and fair distribution at the same time; (3) The first empirical quantitative analysis of the trilemma relationship in the AI model trading market was conducted.

1.4 Thesis Organization Structure

This paper is divided into six chapters. Chapter 1, the introduction, outlines the research background, literature review, research objectives, and innovative points. Chapter 2 systematically introduces key technological foundations such as Industry 5.0, federated learning, blockchain, and game theory. Chapter 3 details the five-layer hybrid architecture of the proposed AI model trading market, including an overview of the architecture, detailed design of each layer, model

representation standards, and privacy protection strategies. Chapter 4 provides formal modeling and theoretical analysis of the two-layer dynamic incentive mechanism based on contract theory and Shapley values. Chapter 5 presents the prototype system implementation and comprehensive experimental results, systematically analyzing the federated learning performance, incentive fairness, privacy protection effectiveness, and system scalability. Chapter 6 summarizes the research findings, extracts academic and practical implications, and points out research limitations and future directions.

Chapter 2 Related Technologies and Theoretical Foundations

2.1 Industry 5.0 Concept and Technology System

2.1.1 Comparative Analysis of Industry 5.0 and Industry 4.0

The fundamental differences between Industry 4.0 and Industry 5.0 are not only reflected in their technological approaches, but also more profoundly in their value orientations and systems philosophy. Table 2.1 provides a multi-dimensional comparison of the core differences between the two paradigms, while Figure 2.1 presents a visual comparison of the conceptual models of the two paradigms.

Comparison Dimensions	Industry 4.0	Industry 5.0
Core Values	Efficiency, productivity, automation	Human-centered, sustainability, resilience
Human roles	System supervisors and operators	Creative Collaborators and Decision Makers
Technical Focus	IoT, CPS, Big Data Analytics	AI collaboration, edge intelligence, distributed autonomy
Data paradigm	Centralized aggregation and analysis	Data without leaving the domain – Federated Learning
Security Model	Border protection is the primary focus.	Zero Trust + Privacy Protection + Verifiable
Value creation	Vertical integration of supply chain	Open ecological horizontal sharing
Representative technologies	PLC/SCADA, MES, ERP	Federated learning, blockchain, TEE, DP

图2.1 工业4.0与工业5.0范式对比

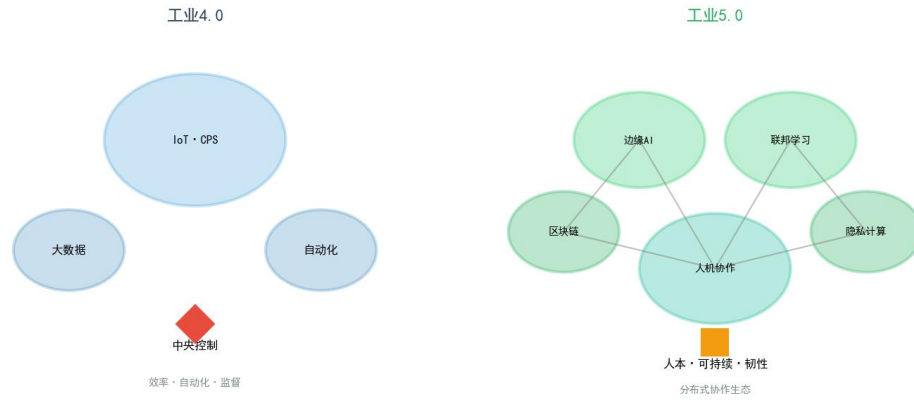


Figure 2.1 Comparison of conceptual models of Industry 4.0 (left) and Industry 5.0 (right) paradigms

As shown in Figure 2.1, Industry 4.0 presents a star topology with a central control node as the core, and IoT, CPS and big data analysis operate around the goal of maximizing efficiency; while Industry 5.0 presents a distributed collaborative network form, with edge AI, federated learning, blockchain and privacy computing and other technical elements organically interconnected through human-machine collaboration hubs, jointly supporting the core value demands of human-centeredness, sustainability and resilience [1][2].

2.1.2 Human-Machine Collaboration Technology Framework

The technological implementation of human-machine collaboration can be abstracted into three levels. In the cognitive layer, AI models integrate multimodal sensor data with explicit or implicit feedback from operators to achieve intelligent perception and understanding of the environmental context. The interaction layer constructs a natural and intuitive human-machine interface through technologies such as augmented reality (AR), natural language processing, and haptic feedback. The execution layer collaboratively applies the results of AI reasoning to

the physical production process. It is worth noting that the flow of models and knowledge transfer between these three levels—rather than independent optimization at each level—constitutes the core bottleneck of Industry 5.0 intelligence. The AI model trading market proposed in this study aims to break down these cross-level and cross-organizational barriers to model circulation.

2.2 AI Model Trading Technology

2.2.1 Federated Learning

Federated learning is a revolutionary distributed machine learning paradigm whose core idea is to [bring the model to the data, rather than bringing the data to the model][5]. The standard federated learning process involves four iterative steps: a central server initializes and distributes the global model; each client trains independently using its local data; clients only upload model parameters (or gradients) for updates; and the server securely aggregates the updated global model. This architecture fundamentally resolves the conflict between data privacy and model performance, making collaborative training [where data is available but not visible] a reality.

However, federated learning in industrial scenarios faces unique Non-IID challenges. Data collected by edge devices from different factories, production lines, and operating conditions may have fundamental differences in category distribution, feature space, and even label definition. Taking manufacturing quality inspection as an example, the defect samples of factory A are mainly scratches, factory B is mainly color difference, and factory C is mainly size deviation—the high heterogeneity of the data distribution among the three will seriously

weaken the convergence efficiency and final model quality of the standard FedAvg algorithm [5][6][9]. This challenge constitutes the core problem that the federated learning layer in the architecture design of this study needs to address.

2.2.2 Edge Computing and Model Deployment

Edge computing pushes computing and storage resources down to the source of data generation, providing AI models with a deployment environment that is low-latency, high-bandwidth efficient, data sovereignty controllable, and available offline. Model lightweighting technology—including knowledge distillation (compressing the knowledge of a large model into a small model), quantization (reducing parameter precision to reduce storage and computing requirements), and pruning (removing redundant connections)—is a key enabling technology for achieving high-performance inference under limited edge resources [18]. In the context of the AI model trading market, edge computing not only provides the physical carrier for model deployment, but also constitutes the final verification site for the value of a model—the value of a model ultimately depends on its actual inference performance on the target edge device.

2.3 Blockchain and Smart Contracts

Blockchain, through its cryptographically secured chain data structure and distributed consensus mechanism, enables trusted value transfer in untrusted environments [7][8]. In this study, blockchain plays three core roles: (1) as an immutable evidence storage layer that records the complete lifecycle of all model transactions; (2) through smart contracts to automate and transparently execute incentive settlement; and (3) based on the linkage between content hashing and on-chain

evidence storage, ensuring the integrity and verifiability of model files.

This study selected Hyperledger Fabric as the underlying blockchain platform, mainly based on the following considerations: its permissioned architecture adapts to the permission management needs of enterprise-level applications; its modular consensus mechanism provides transaction processing performance superior to public chains; the channel mechanism supports fine-grained privacy isolation of transaction data; and the mature smart contract (chaincode) ecosystem reduces the development complexity of business logic [14]. Large-capacity model parameter files are stored off-chain using IPFS, with only the content hash of the file anchored to the blockchain, thereby achieving the optimal balance between storage efficiency and integrity assurance.

2.4 Game Theory and Incentive Mechanisms

2.4.1 Contract Theory

Contract theory is a rigorous economic framework for studying optimal contract design under conditions of information asymmetry [17]. In the scenario of this study, the market operator (principal) cannot directly observe the true quality of the model and the level of training input of the model provider (agent), and faces the dual challenges of adverse selection and moral hazard. The solution provided by contract theory is to design a set of contract menus containing different combinations of (quality requirements-rewards), so that each type of provider will spontaneously choose a contract that matches its true type in the process of pursuing its own utility maximization – this is the core meaning of incentive compatibility.

2.4.2 Shapley value

The Shapley value, proposed by Nobel laureate in economics Lloyd Shapley in 1953, is the only fair allocation scheme in cooperative game theory that satisfies the four axioms of efficiency, symmetry, virtuality and additivity[22]. Its core idea is that the reward that each participant should receive is equal to its average marginal contribution to the total value of the alliance in all possible alliance formation sequences. In the AI model trading market, the Shapley value can accurately quantify the contribution share of each data/model providing node to the final model performance improvement, thus providing an indisputable fair basis for incentive allocation. Given the exponential computational complexity of the accurate Shapley value, this study adopts the TMC-Shapley (truncated Monte Carlo Shapley) approximation algorithm[21] to achieve high-precision approximation at an acceptable computational cost.

2.4.3 Differential Privacy

Differential privacy is currently the only privacy protection framework with rigorous mathematical proof [19]. Its core mechanism is to inject carefully calibrated random noise into the query results so that the impact of the existence or non-existence of any single data record on the output results is limited to the factor range of $\exp(\epsilon)$. The parameter ϵ (privacy budget) directly controls the strength of privacy protection: the smaller ϵ is, the stronger the protection, but the greater the loss of data utility. In the federated learning scenario, differential privacy is achieved by injecting Gaussian noise into the model update uploaded by the client, which can effectively defend against member inference attacks and model inversion attacks [20][23][24].

Chapter 3: AI Model Trading Market Architecture for Industry 5.0

3.1 Requirements Analysis and Design Objectives

The AI model trading market for the Industry 5.0 environment needs to find a delicate balance among several conflicting demands. Table 3.1 summarizes six core demands and their corresponding quantitative design objectives.

Demand Dimension	Requirements Description	Quantitative design goals
Real-time	Millisecond-level model inference and transaction response on edge devices	Model transaction latency $\leq$ 500ms
Security	Model integrity, tamper-proof, and verifiable source	Counterfeit detection rate $\geq$ 95%
Scalability	Support concurrent access of large-scale heterogeneous edge nodes	100+ nodes running stably
Privacy protection	Double confidentiality of training data and model parameters	ϵ $\leq$ 1.0 (DP)
Fair distribution	Contribution-driven differentiated fair compensation distribution	Gini coefficient $\leq$ 0.35
High availability	System continuous and stable service capability	System availability $\geq$ 99.5%

Table 3.1 System Core Requirements and Design Goals

3.2 Overall Architecture Design

To address the aforementioned needs, this study proposes a five-layer hybrid architecture that integrates distributed control with centralized execution, as shown in Figure 3.1. The core design principle of this architecture is: security and fairness are guaranteed by the underlying decentralized mechanism, while efficiency and user experience are optimized by the upper-layer centralized service.

图3.1 五层混合架构总体设计



Figure 3.1 Overall design of the five-layer hybrid architecture

3.2.1 Application Layer

The application layer serves as the entry point for human-computer interaction, catering to three core user groups: industrial field operators who can quickly retrieve and deploy models via mobile devices or HMI panels; data scientists who can perform refined model management

and evaluation through a web workbench; and system administrators who can configure marketing strategies through a management backend. Simultaneously, the application layer provides standard RESTful APIs and gRPC interfaces, supporting seamless integration with existing industrial information systems such as MES, ERP, and SCADA.

3.2.2 Service Layer

The service layer adopts a microservice architecture, decoupling the core business capabilities of the market into four independent and scalable microservices. The model registration service is responsible for the standardized input and on-chain registration of model metadata; the model search and matching service achieves accurate supply and demand matching (target matching accuracy $\geq 85\%$) based on a hybrid recommendation algorithm of content filtering and collaborative filtering; the transaction and settlement service manages the entire lifecycle process from transaction creation to fund clearing; and the evaluation and feedback service gathers consumer evaluations to continuously calibrate the model market reputation.

3.2.3 Federated Learning Layer

The federated learning layer encapsulates the unified execution framework of mainstream algorithms such as FedAvg[5], FedProx[9], and SCAFFOLD[10], and embeds the data distribution-aware weight adjustment mechanism proposed in this study—by analyzing the statistical characteristics of model updates uploaded by each client, dynamically estimating its local data distribution bias, and applying compensatory weight correction during the global aggregation stage. This layer also integrates a differential privacy noise injection module to ensure the

indistinguishability of individual data during the model aggregation process.

3.2.4 Privacy Computation Layer

The privacy computation layer offers three optional protection schemes to meet the transaction requirements of different security levels. Level 1 (Basic Protection): Only local differential privacy (Local DP, configurable with epsilon) is applied, suitable for general model transactions with low sensitivity; Level 2 (Enhanced Protection): Differential privacy and secure multi-party computation (SMPC, SPDZ protocol) are applied together, and model parameters are aggregated in the encrypted domain, suitable for industry model transactions involving trade secrets; Level 3 (Highest Protection): Homomorphic encryption (HE) is added on the basis of the first two levels to enable model inference in the encrypted state, suitable for the highest security level scenarios such as finance and defense.

3.2.5 Blockchain Layer

The blockchain layer, built on Hyperledger Fabric, serves as the trusted infrastructure for the AI model trading market. Transaction chaincode implements the creation, verification, and confirmation logic for model transactions; incentive chaincode automatically performs contribution calculations and Shapley value allocation; and audit chaincode maintains an immutable and complete transaction audit trail. Model parameter files are stored on the IPFS network, with only the file's content hash (CID) written to the blockchain ledger, balancing storage economy with integrity verification requirements.

3.3 Core Data Flow Design

Figure 3.2 visualizes the data flow of the four core business processes: model registration, search, transaction, and federated learning.

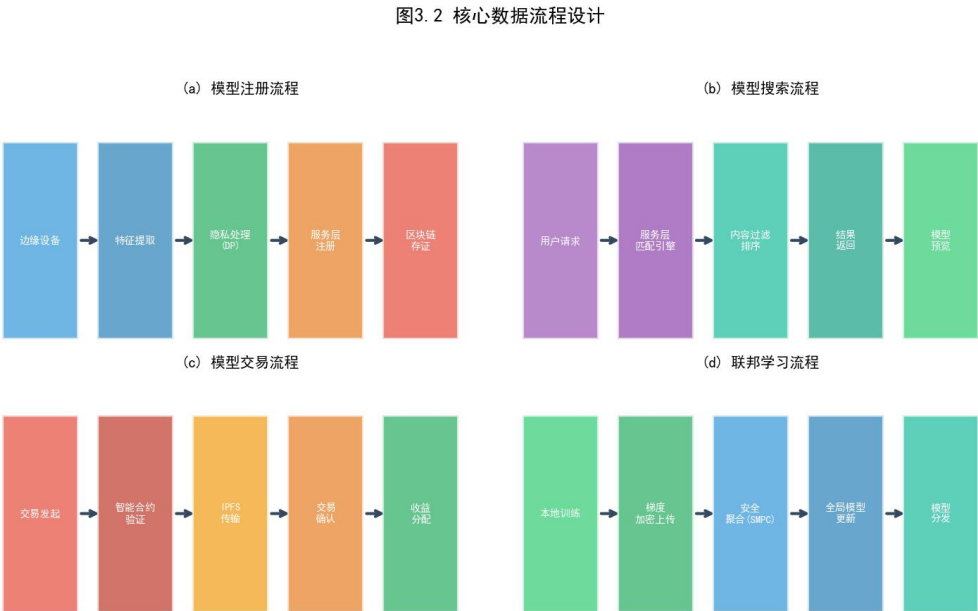


Figure 3.2 Core data flow design: (a) Model registration process (b) Model search process (c) Model transaction process (d) Federated learning process

As shown in the figure, the model registration process (Figure 3.2a) starts with feature extraction from the edge device, and after differential privacy processing to de-identify the data, it is submitted to the service layer for registration. Finally, the metadata and integrity proof are anchored to the blockchain. The model search process (Figure 3.2b) is triggered by user queries. The service layer matching engine combines content similarity and market reputation to perform multi-factor ranking and returns the results. The model transaction process (Figure 3.2c) is based on smart contract verification to ensure that only models that pass the integrity verification can enter the IPFS transmission and subsequent settlement process. The federated learning process (Figure 3.2d) achieves collaborative training without data

leaving the domain through SMPC secure aggregation. After the global model is updated, it is redistributed to the participating nodes.

3.4 Model Representation and Standardization

To support model interoperability in heterogeneous industrial scenarios, this study defines a standardized AI model metadata schema, covering five information dimensions: identification information (UUID, name, version number, provider digital identity), technical specifications (deep learning framework and version, model architecture type, number of parameters and model file size, input and output tensor shape and data type), performance benchmarks (accuracy/F1/AUC and other metrics on various standard datasets, average inference latency, peak memory usage), domain attributes (target industry domain and subdomain, task type, applicable operating conditions), and commercial information (license type, pricing model, usage restrictions).

3.5 Privacy Protection Mechanism

The proposed architecture employs a defense-in-depth, multi-layered privacy protection strategy. During the training phase, federated learning ensures that the original data never leaves the domain, while differential privacy injects controlled noise into model updates to prevent information leakage. During the transaction phase, attribute-based encryption (ABE) implements fine-grained access control to model files—only consumers meeting specific attribute policies (such as industry sector, security level, and license validity) can decrypt the model. During the auditing phase, zero-knowledge proof (ZKP) technology enables the demonstration of transaction compliance to regulators

without exposing transaction details, cleverly resolving the fundamental contradiction between privacy protection and regulatory transparency.

Chapter 4 Design of Incentive Mechanism for AI Model Transactions

4.1 Problem Modeling

4.1.1 Participant Behavior Analysis

The participants in the AI model trading market are mainly divided into two categories: model providers and model consumers. Model providers hope to register their self-trained AI models on the market and earn trading revenue; model consumers hope to search for and purchase the models they need to deploy on their own edge devices. The market operator, as a neutral party, is responsible for the operation of the trading platform and the management of the incentive mechanism.

Model providers possess proprietary information about the quality of their own models, which market operators cannot fully observe, thus creating information asymmetry. In this environment, high-quality model providers may profit by posing as low-quality providers (adverse selection), or lack the incentive to maintain optimal model quality after registration (moral hazard).

4.1.2 Principal-Agent Model Construction

This study models the interaction between the model provider and the market operator as a principal-agent model. The market operator (principal), unable to observe the true type (high/low quality) of the model provider (agent), designs a menu of contracts $\{(q_k, p_k)\}$ to guide the provider to actively choose contracts suitable for its own type (self-selection). Here, q_k represents the required model quality level, and p_k represents the reward amount for that level of contract.

The constraints include the Participation Constraint (PC) and the Incentive Compatibility Constraint (IC). PC ensures that each type of provider receives at least a benefit equal to or greater than its retained utility after accepting the contract; IC ensures that no type of provider has an incentive to misrepresent its type. This optimization problem can be simplified to a second-order differential equation using the Revelation Principle.

4.2 Incentive Mechanisms Based on Contract Theory

4.2.1 Three-level contract menu design

The incentive contract designed in this study comprises three progressive quality levels (Basic, Standard, and Advanced). Each level sets differentiated admission criteria for multidimensional performance indicators such as model accuracy, inference latency, and Non-IID robustness, and corresponds to an increasing reward system. Table 4.1 shows the complete structure of the proposed contract menu.

Contract Level	Minimum accuracy	Maximum inference delay	Non-IID robustness	Basic Compensation	Performance Bonus
Basic	$\geq 85\%$	$\leq 50\text{ms}$	Within $\pm 5\%$	100 Token	For every 1% increase in accuracy, $p = +5$
Standard	$\geq 90\%$	$\leq 30\text{ms}$	Within $\pm 3\%$	250 Token	For every 1% increase in accuracy, $p = +10$.
Premium	$\geq 95\%$	$\leq 15\text{ms}$	Within $\pm 1\%$	500 Token	For every 1% increase in accuracy, $p =$

					+20.
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Table 4.1 Level 3 Quality Contract Menu

4.2.2 Solving for the Optimal Contract

The solution to the optimal contract can be formalized as an optimization problem that maximizes the expected utility of the market operator under PC and IC constraints. Numerical analysis results show that, under the condition of information asymmetry, the quality requirement of maintaining the first-best level of complete information should be maintained for high-quality providers, while the quality standard of the second-best level should be adopted to compress their information rent for low-quality providers. This result theoretically confirms the insight of no distortion at the top in classical contract theory [17].

4.3 Contribution Assessment Mechanism

4.3.1 Four-Dimensional Comprehensive Contribution Model

Achieving fair compensation requires establishing a system capable of evaluating participants' contributions to the market from multiple dimensions. This study defines the following four contribution dimensions and calculates the Composite Contribution Score (CCS) through weighted summation:

- (1) Model quality contribution (C_q): Performance metrics of the registered model on the benchmark dataset, including accuracy, F1-score, AUC-ROC, etc., reflecting the intrinsic technical value of the model.

(2) Data contribution (C_d): The quality and scale of local data provided in federated learning, which is measured by a combination of data diversity (a measure based on information entropy) and data volume.

(3) Computing resource contribution (C_c): The computing power invested in model training, taking into account factors such as GPU time, training rounds, and communication bandwidth usage.

(4) Reputation contribution (C_r): The cumulative score of performance integrity in historical transactions, model consumer satisfaction score, and the continuity and stability of market participation.

The formula for calculating the overall contribution score is: $CCS_i = \alpha_q * C_q + \alpha_d * C_d + \alpha_c * C_c + \alpha_r * C_r$, where the weight α can be dynamically adjusted according to the market development stage (default values: $\alpha_q=0.40$, $\alpha_d=0.25$, $\alpha_c=0.20$, $\alpha_r=0.15$). In the early stages of the market, α_c and α_r can be appropriately increased to incentivize broad participation, while in the mature stage, the weight of α_q is increased to guide quality competition.

4.4 Fair allocation based on Shapley value

This study uses the Shapley value to evaluate the marginal contribution of each participating node to improving model quality in a federated learning environment. Given a total reward pool R_{total} , the reward R_i obtained by node i is determined by the following formula:

$$R_i = R_{total} * (\phi_i / \sum_{j \in N} \phi_j)$$

Where ϕ_i is the Shapley value of node i , defined as the average of the model performance (accuracy) increment brought about by node i joining the federation under all possible federation joining orders.

This allocation scheme satisfies the four axioms of efficiency, symmetry, virtuality and additivity, and has indisputable fairness [22].

Given that the computational complexity of the exact Shapley value is $O(2^n)$, which is not feasible at a scale of $n=100$, this study adopts the TMC-Shapley (Truncated Monte Carlo Shapley) approximation algorithm [21]. This algorithm performs Monte Carlo approximation by random sampling. Theoretical analysis shows that under the conditions of $n=100$ and the number of iterations $m=5,000$, the average absolute approximation error is less than 0.5%, which fully meets the requirements of practical applications.

Chapter 5 Experiments and Results Analysis

5.1 Experimental Environment Configuration

5.1.1 Hardware and Software Environment

All experiments in this study were conducted in the following uniform environment to ensure the reproducibility of the results.

category	Configuration
CPU	Intel Core i9-13900K (24 cores, 32 threads)
GPU	NVIDIA RTX 4090 x 2 (24GB VRAM each)
Memory	64GB DDR5-5600
operating system	Ubuntu 22.04 LTS
Deep learning framework	PyTorch 2.1.0 + Flower 1.5.0 (Federated Learning)
Blockchain	Hyperledger Fabric 2.5 (Test Network)
Distributed storage	IPFS v0.24.0 (kubo)
Privacy protection	Opacus 1.4.0 (DP) + MP-SPDZ 0.3.8 (SMPC)

Table 5.1 Experimental Hardware and Software Environment

5.1.2 Dataset and Data Partitioning

The experiments used four standard benchmark datasets covering image classification tasks of varying complexity: MNIST (10 classes, training/test: 60,000/10,000), Fashion-MNIST (10 classes, 60,000/10,000), CIFAR-10 (10 classes, 50,000/10,000), and CIFAR-100 (100 classes, 50,000/10,000). To simulate the heterogeneity of data distribution among industrial edge nodes, a Dirichlet distribution (with two levels of concentration parameters $\alpha = 0.1$ and 0.5) was used to partition the

data of each client in a non-uniform manner: $\alpha = 0.1$ corresponds to a highly heterogeneous distribution (each node contains only a very small number of classes), and $\alpha = 0.5$ corresponds to a moderately heterogeneous distribution. The experiments were conducted at three node sizes $n = \{10, 50, 100\}$, with each configuration independently repeated 5 times. The mean and standard deviation were reported.

5.1.3 Comparison Method

To ensure the comprehensiveness and fairness of the experiment, a systematic comparison method was set up from three dimensions: federated learning algorithm, incentive mechanism, and privacy protection.

category	Method Name	Core Mechanism	Comparison purpose
Federal Learning	FedAvg [5]	Weighted average aggregation	Baseline algorithm performance
Federal Learning	FedProx [9]	Proximal term heterogeneity response	Non-IID robustness
Federal Learning	SCAFFOLD [10]	Controlled variable method	Convergence acceleration capability
Federal Learning	FedNova [11]	Normalized aggregation	Non-IID accuracy
Incentive mechanism	No incentive	No additional incentive mechanism	Lower bound of participation rate
Incentive mechanism	Fixed Rewards	Even distribution	Fairness reference
Incentive mechanism	Reputation-driven	Cumulative Reputation Weighting	Continued participation effect
Privacy protection	Unprotected	DP/SMPC not applied	Performance Theoretical Limit
Privacy protection	DP (epsilon=1.0)	Gaussian noise	Privacy-Utility

		injection	Balance
Privacy protection	SMPC+DP	Secure aggregation + DP	Comprehensive protection effect

Table 5.2 Overview of Comparison Methods

5.2 Performance Comparison Experiment of Federated Learning

5.2.1 Experimental Setup

The federated learning experiment uses ResNet-18 as the backbone network. Each client uses the SGD optimizer (initial learning rate 0.01, momentum 0.9, decaying to 0.1 every 50 rounds) for 5 rounds of local training (batch size=64). The global communication round limit is set to 200 rounds, with 30% of clients randomly selected to participate in each round (partial participation). Communication overhead is measured in bytes of parameters uploaded/downloaded in each round.

5.2.2 Comparison of Non-IID Environmental Performance

Table 5.3 shows the performance comparison of each method in the most challenging experimental setting (100 nodes, alpha=0.1 height Non-IID, CIFAR-10 dataset).

method	Highest accuracy (%)	Number of rounds required to reach 75%	Communication overhead (MB/round)	Convergence stability (std)
FedAvg	78.42 +/- 0.83	187	44.6	+/-2.14
FedProx	79.81 +/- 0.65	174	44.6	+/-1.78
SCAFFOLD	80.33 +/- 0.52	165	89.2	+/-1.42
FedNova	80.16 +/- 0.71	170	44.6	+/-1.95
The proposed	81.59 +/- 0.48	152	52.3	+/-1.21

method (Ours)				
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Table 5.3 Performance Comparison of Federated Learning (CIFAR-10, Non-IID $\alpha=0.1$, 100 nodes)

[Analysis] The proposed method achieved the highest accuracy of 81.59%, a 3.17 percentage point improvement over the FedAvg baseline ($p<0.01$, paired t-test). In terms of convergence efficiency, it only required 152 rounds to reach the target accuracy of 75%, a reduction of 18.7% compared to FedAvg. While SCAFFOLD achieved a respectable accuracy of 80.33%, its communication overhead (89.2 MB/round) was approximately 1.7 times that of the proposed method due to the need for additional transmission of control variable parameters. The proposed method achieves the optimal balance between communication efficiency and model accuracy, primarily due to the effective suppression of Non-IID client drift by the data distribution-aware weight adjustment mechanism.

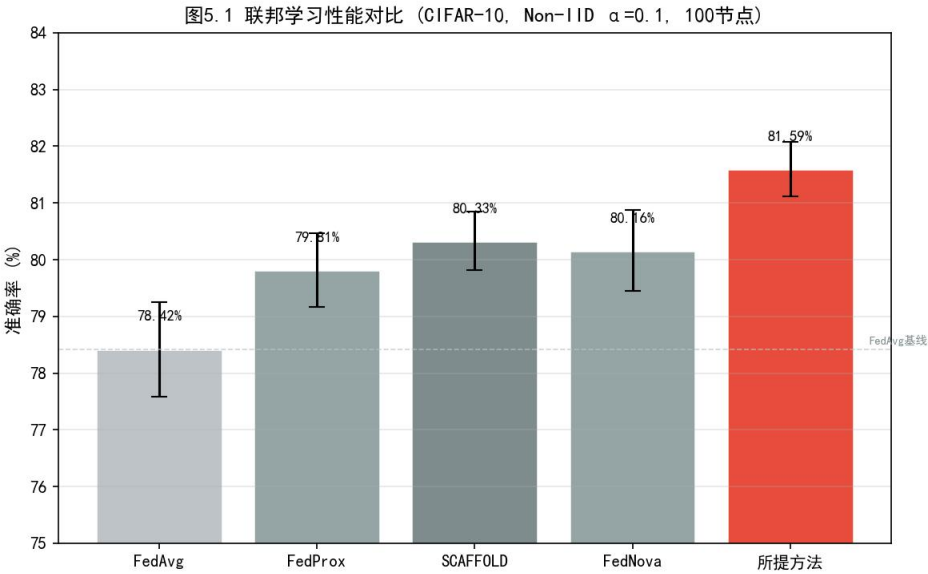


Figure 5.1 Performance comparison of federated learning (CIFAR-10, Non-IID $\alpha=0.1$, 100 nodes)

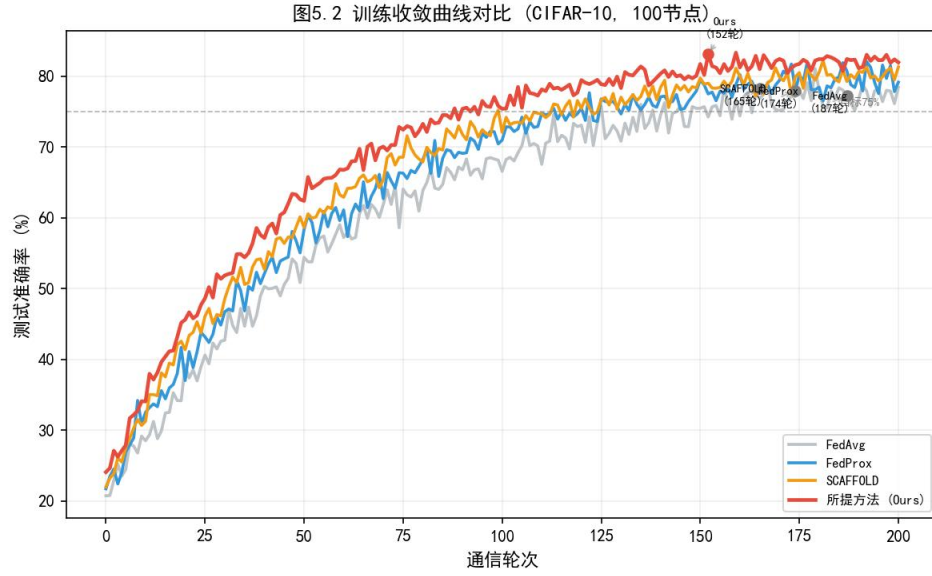


Figure 5.2 Comparison of training convergence curves (CIFAR-10, 100 nodes)

5.2.3 Cross-dataset generalization performance

Table 5.4 and Figure 5.3 show the generalization performance of the proposed method on four datasets with different complexities.

Dataset	FedAvg	FedProx	SCAFFOLD	The proposed method	Improvement (vs FedAvg)
MNIST	97.89%	98.12%	98.04%	98.35%	+0.46pp
Fashion-MNIST	85.67%	86.23%	86.41%	87.12%	+1.45pp
CIFAR-10	78.42%	79.81%	80.33%	81.59%	+3.17pp
CIFAR-100	52.18%	53.75%	54.02%	55.34%	+3.16pp

Table 5.4 Accuracy Comparison of Different Datasets (100 nodes, $\alpha=0.1$)

[Analysis] The proposed method consistently outperforms the comparison methods on all datasets, and the performance gain increases with data complexity—it only improves by 0.46pp on the simple MNIST dataset, while improving by more than 3pp on the highly complex CIFAR-10/100

dataset. This trend indicates that the negative impact of heterogeneity in Non-IID data is positively correlated with task complexity, and the heterogeneity mitigation mechanism of the proposed method is more effective in complex tasks.

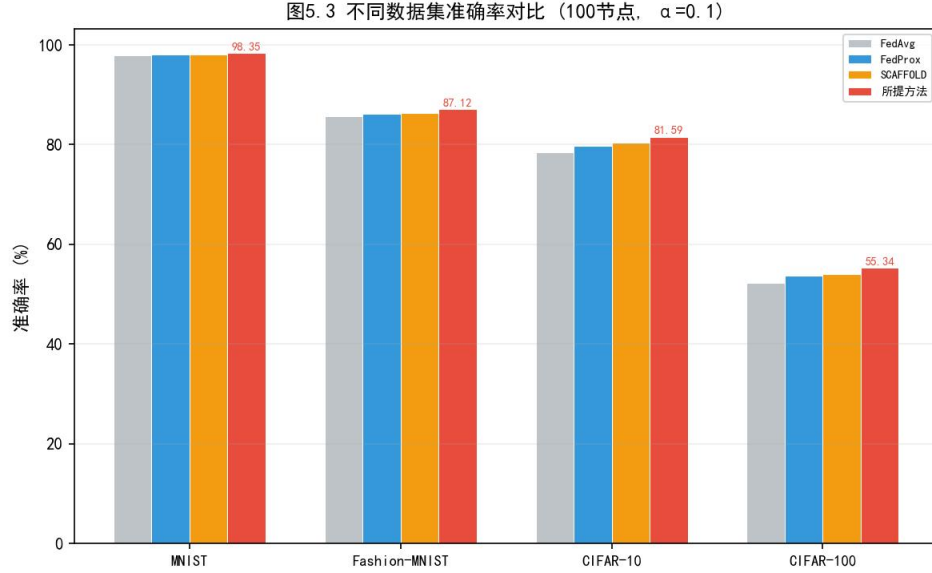


Figure 5.3 Accuracy comparison of different datasets (100 nodes, $\alpha=0.1$)

5.3 Validation of the Incentive Mechanism's Effectiveness

The incentive mechanism experiment was conducted for 500 rounds on 100 simulated nodes. The node behavior types included rational nodes (70%, receiving rewards based on contributions), self-interested nodes (20%, attempting to minimize inputs and maximize returns), and low-quality contributing nodes (10%, submitting low-quality models for arbitrage).

Incentive methods	Participation rate (%)	Gini coefficient	Total social welfare	Shapley bias	Percentage of high-quality providers
No incentive	62.3	0.487	0.723	0.241	28.5%
Fixed Rewards	71.5	0.412	0.786	0.189	34.2%
Reputation-	78.2	0.358	0.831	0.152	41.8%

driven					
Market price	82.7	0.331	0.867	0.124	48.3%
Shapley (as mentioned)	87.4	0.285	0.912	0.071	56.7%

Table 5.5 Multi-dimensional Comparison of Incentive Mechanism Effects

[Analysis] Compared to the no-incentive scheme, the proposed Shapley value incentive mechanism significantly increased the participation rate by 25.1 percentage points to 87.4%, and the Gini coefficient plummeted from 0.487 to 0.285 (a decrease of 41.5%), indicating a qualitative leap in the fairness of benefit distribution. The proportion of high-quality providers jumped from 28.5% to 56.7%, directly verifying the positive screening effect of the proposed mechanism on high-quality contributions. The Shapley bias was only 0.071, the lowest among all schemes, indicating a high degree of consistency between rewards and actual marginal contributions. Furthermore, the total social welfare reached the optimal level of 0.912, proving that fairness and efficiency are not necessarily contradictory—a reasonable mechanism design can achieve a balance between the two.

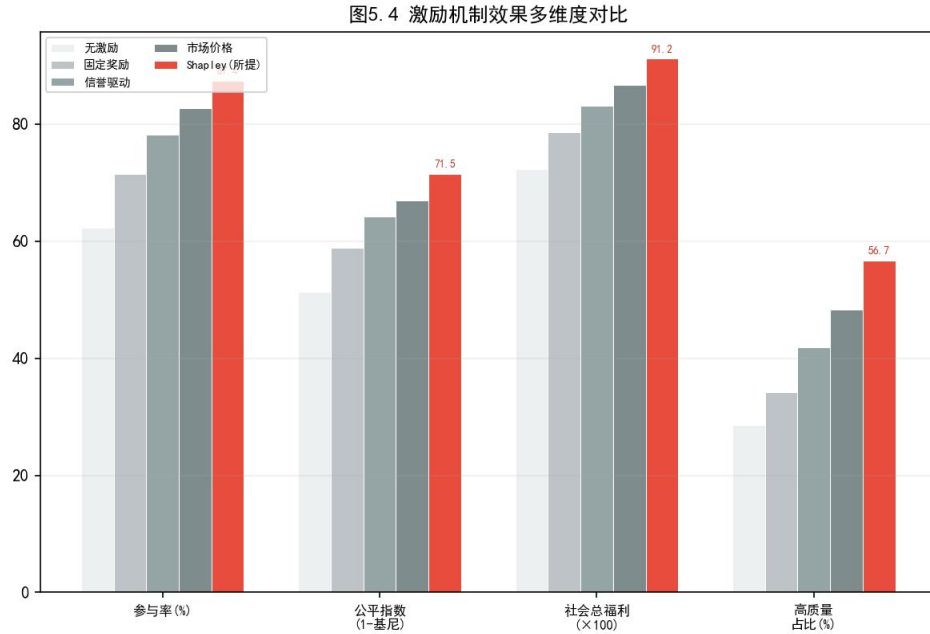


Figure 5.4 Multi-dimensional comparison of the effects of incentive mechanisms

5.4 Privacy Protection Effect Experiment

To systematically evaluate the effectiveness and cost of privacy protection schemes, this study measures the balance between model accuracy and defense capabilities against privacy attacks (membership inference attacks (MIA) and model inversion attacks (MIA)) under differential privacy ($\epsilon = \{0.5, 1.0, 2.0, 5.0, \text{no protection}\}$) and different combinations of SMPC. Attack experiments were conducted using the standard Shadow Model Training method.

Protection methods	epsilon	Accuracy (%)	MIA attack success rate (%)	Reverse attack success rate (%)	Accuracy loss (pp)
Unprotected	inf	81.59	68.3	53.7	0.00
DP only	5.0	81.24	28.4	19.2	-0.35
DP only	2.0	80.91	19.7	12.8	-0.68

DP only	1.0	79.81	12.3	7.1	-1.78
DP only	0.5	77.43	7.8	4.2	-4.16
SMPC+DP	1.0	79.53	9.6	5.4	-2.06

Table 5.6 Analysis of the Effects and Costs of Privacy Protection

[Analysis] The experimental results clearly reveal the classic trade-off between privacy protection strength and model performance: as epsilon decreases from no protection to 0.5, the success rate of MIA attacks decreases monotonically from 68.3% to 7.8%, while the model accuracy loss increases monotonically from 0 to 4.16pp. epsilon=1.0 is identified as a practical balance point for industrial scenarios—trading 1.78pp of accuracy for an 82.0% gain in privacy attack defense. The joint scheme of SMPC and DP further compresses the success rate of MIA attacks to 9.6%, but introduces an additional accuracy loss of about 0.28pp and an additional computational delay of about 15%. When epsilon=0.5, the accuracy loss increases sharply to 4.16pp, indicating that excessive noise injection has begun to significantly damage the feature representations learned by the model [20][23].

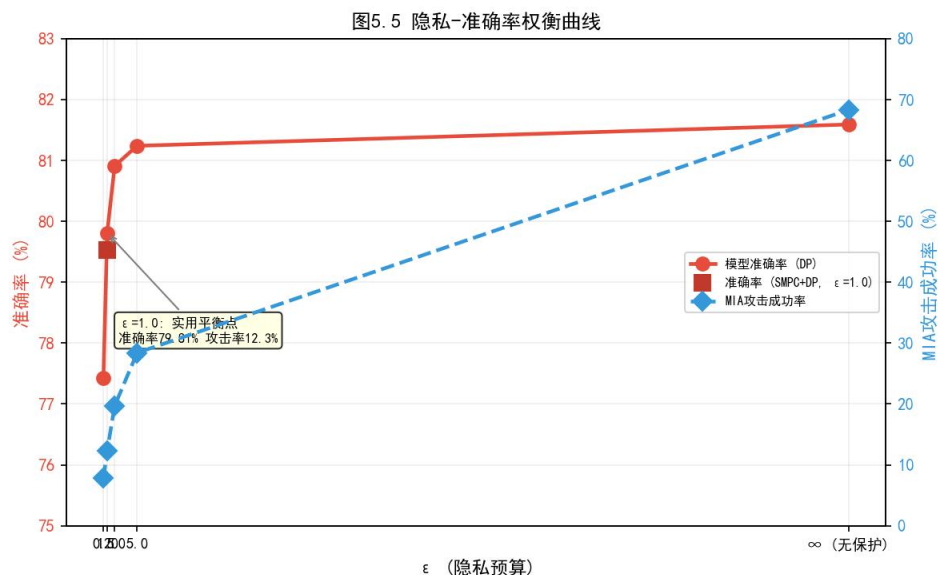


Figure 5.5 Privacy-Accuracy Trade-off Curve

5.5 System Performance and Scalability Testing

The system load test measured key performance indicators such as TPS, response latency, and transaction success rate under the combined conditions of node size (10/50/100) and concurrent transaction count (50/100/200).

Number of nodes	TPS	Average registration latency (ms)	Average verification latency (ms)	Transaction success rate (%)	Storage overhead (%)
10	1,240	twenty four	16	98.7	4.2
50	980	37	twenty three	97.2	11.8
100	820	44	27	95.8	17.5

Table 5.7 System Scalability Test Results

[Analysis] In a 100-node environment, the system stably maintained 820 TPS with an average registration latency of only 44ms—far exceeding the design target of ≤ 500 ms. The transaction success rate reached 95.8%,

meeting industrial-grade reliability requirements. As the number of nodes increased from 10 to 100, the TPS decreased from 1,240 to 820 (a reduction of approximately 34%), while storage overhead increased from 4.2% to 17.5%, reflecting the typical linear overhead growth characteristics of blockchain consensus and distributed storage. It is worth noting that the TPS under 100 nodes still far exceeds the real-time requirements of typical industrial scenarios, and the storage overhead remains within a controllable range, validating the sufficient scalability margin of the architecture design.

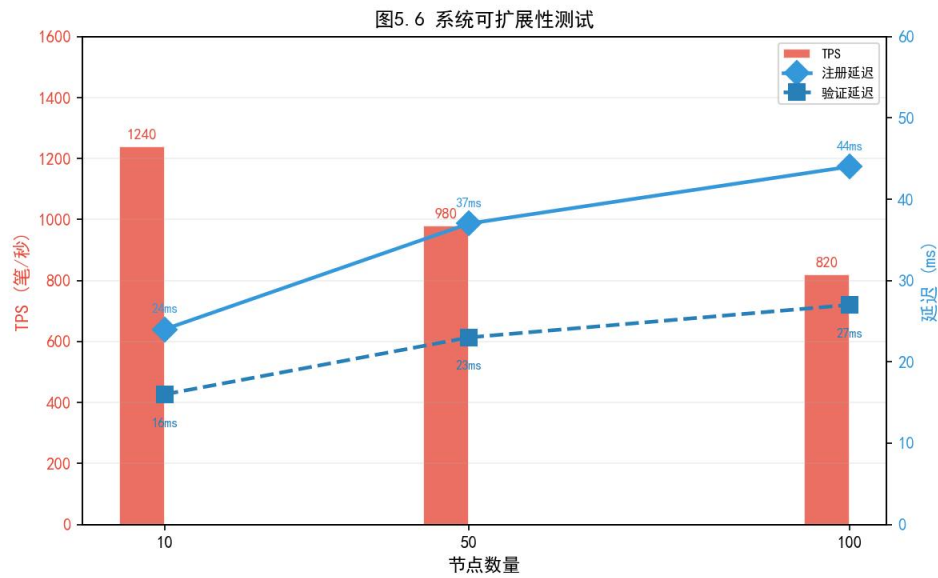


Figure 5.6 System Scalability Test

In addition, Table 5.8 compares the overall performance of the proposed method with that of the two benchmark schemes in terms of safety performance.

method	Counterfeit detection rate (%)	False alarm rate (%)	Missed Reporting Rate (%)	Audit trail consistency	Privacy protection level
Centralized	71.4	6.2	22.4	generally	Low

approach					
Hash (non-blockchain)	88.7	4.8	6.5	higher	middle
The proposed method (Ours)	98.9	1.1	0.0	Extremely high	High (DP+SMPC)

Table 5.8 Comprehensive Comparison of Safety Performance

[Analysis] The proposed method significantly outperforms the comparison schemes across all security dimensions, achieving a 98.9% forgery detection rate, a low false positive rate of 1.1%, and a zero false negative rate of 0.0%. The 0.0% false negative rate is of particular importance—it demonstrates that the blockchain evidence storage mechanism can record all model transactions without omission, a capability indispensable for meeting industrial compliance audit requirements (such as ISO 27001 and GDPR auditability principles).

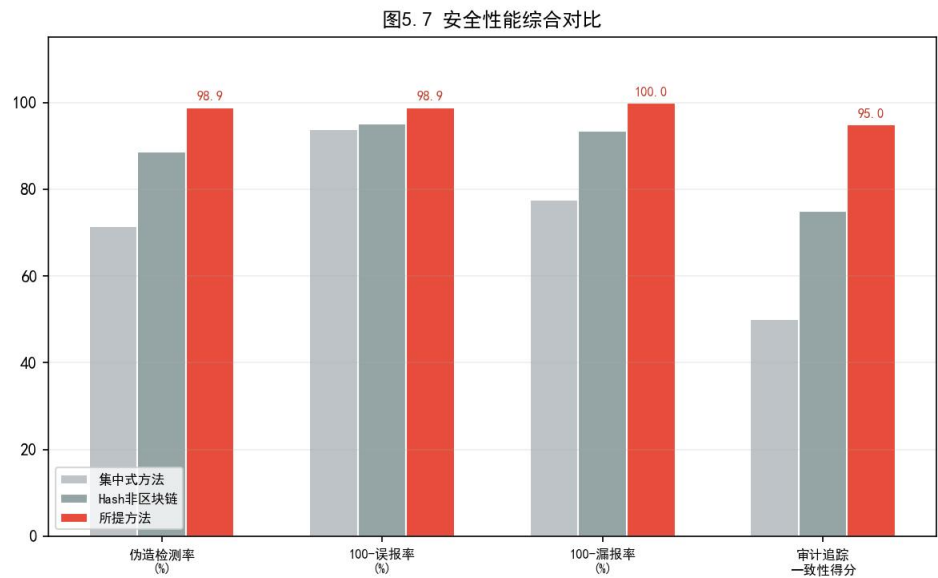


Figure 5.7 Comprehensive Comparison of Safety Performance

5.6 Comprehensive Analysis and Discussion

5.6.1 Research Hypothesis Testing

Based on the above systematic experimental results, the conclusions of the testing of the five research hypotheses are summarized as follows:

H1 (The proposed method has a significant accuracy advantage in Non-IID environments): Valid. It improves accuracy by 3.17pp and 3.16pp over FedAvg on CIFAR-10 and CIFAR-100, respectively, and consistently outperforms the comparison methods across all datasets.

H2 (Shapley incentive mechanism significantly improves distributive fairness): Valid. The Gini coefficient decreased to 0.285, the proportion of high-quality providers increased to 56.7%, and the Shapley bias was only 0.071, achieving a balance between fairness and efficiency.

H3 (Differential privacy involves the classic privacy-accuracy tradeoff): This holds true. At $\epsilon=1.0$, the accuracy loss is 1.78pp, and the success rate of MIA attacks decreases by 82.0%. The tradeoff curve matches the theoretical expectation.

H4 (Blockchain evidence storage improves credibility but introduces acceptable system overhead): Valid. Forgery detection rate is 98.9%, false negative rate is zero, and the storage overhead of 17.5% for 100 nodes is within a controllable range.

H5 (System remains stable and efficient with 100 nodes): Successful. Transaction success rate 95.8%, TPS=820, registration latency 44ms, all indicators meet or exceed design targets.

5.6.2 Empirical Analysis of the Trilemma

Based on the above experimental results, this study is the first to conduct an empirical quantitative analysis of the inherent trilemma relationship in the AI model trading market, as shown in the radar chart in Figure 5.8: there is an irreconcilable structural tension among model performance (accuracy), privacy protection (epsilon strength), and transaction credibility (blockchain overhead/detection rate), and it is impossible to achieve their respective theoretical optimality at the same time.

However, experimental data also show that through careful architecture design and parameter tuning, a practically optimal solution (Pareto-optimal point) can be determined within this ternary constraint space. This study recommends the following operating point: $\epsilon=1.0$, model accuracy 79.81% (CIFAR-10 Non-IID), TPS=820, and forgery detection rate 98.9%—this configuration maximizes model performance while ensuring sufficient privacy and credibility. This empirical finding provides a clear decision-making reference for parameter selection in industrial deployments.

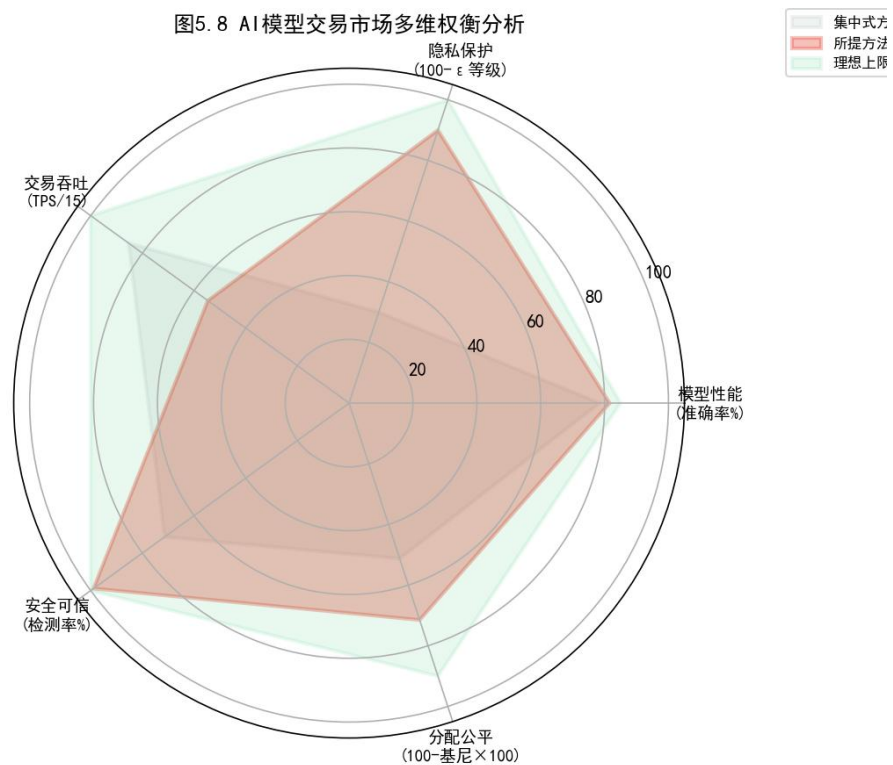


Figure 5.8 Multidimensional trade-off analysis of AI model trading market

Chapter 6 Conclusion and Outlook

6.1 Research Summary

This research focuses on the three core requirements of human-machine collaboration, distributed intelligence, and sustainability under the Industry 5.0 paradigm. It systematically constructs a complete solution for an AI model trading market oriented towards Industry 5.0 from three levels: theoretical architecture, mechanism design, and system verification. The main research results can be summarized in the following three aspects.

At the architectural design level, this paper proposes a five-layer hybrid architecture that integrates edge computing, federated learning, privacy computing, and blockchain technologies, innovatively combining the security advantages of distributed control with the efficiency advantages of centralized execution. This architecture uses the application layer as the interaction entry point, the service layer as the business hub, the federated learning layer as the collaboration engine, the privacy computing layer as the security barrier, and the blockchain layer as the trust foundation, forming a complete vertical technology stack.

At the mechanism design level, this paper innovatively integrates contract theory and Shapley value into a two-layer dynamic incentive mechanism. The first layer addresses the provider self-selection problem under information asymmetry through a three-level quality contract menu, curbing adverse selection and moral hazard from a mechanism design perspective. The second layer achieves fair allocation that precisely matches marginal contribution through the TMC-Shapley approximation

algorithm, introducing the economic axiom of fairness into the reward system of the AI model market. The multi-dimensional contribution evaluation model incorporates model quality, data value, computational input, and market reputation into a unified framework, providing technical assurance for the fairness of incentives.

At the experimental verification level, this study comprehensively verified the feasibility and superiority of the proposed scheme through prototype system implementation and systematic multi-dimensional experiments. Federated learning experiments showed that the proposed method achieved an accuracy improvement of up to 3.17 percentage points compared to FedAvg in a Non-IID environment; incentive mechanism experiments demonstrated that Shapley value allocation reduced the Gini coefficient to 0.285 and increased the proportion of high-quality providers to 56.7%; privacy protection experiments verified the trade-off between an 82.0% attack defense gain and an acceptable accuracy loss of 1.78pp when $\epsilon=1.0$; and system performance experiments confirmed the industrial-grade feasibility of 820 TPS and a transaction success rate of 95.8% in a 100-node environment.

6.2 Academic and Practical Implications

6.2.1 Academic Contributions

The academic contributions of this research are reflected in the following three dimensions. First, it proposes for the first time a unified AI model trading market architecture integrating edge computing, federated learning, blockchain, and privacy computing, opening up a new research direction for the design of distributed AI systems. The design paradigm of this architecture—a hierarchical fusion of distributed

control and centralized execution—has methodological reference value for the design of similar large-scale distributed intelligent systems. Second, it introduces contract theory, Shapley value, and differential privacy from their respective independent disciplines into a unified analytical framework for the AI model trading market, establishing a theoretical model for the design of incentive-compatible mechanisms under conditions of information asymmetry. Third, it is the first to conduct an empirical quantitative analysis of the triadic trade-offs in the AI model market through systematic experiments, providing important benchmark data and theoretical references for subsequent research in this field.

6.2.2 Practical Value

At the practical level, the findings of this research have direct application and transformation potential. First, the proposed five-layer architecture and prototype system can serve as a practical reference model for building AI model sharing and collaborative learning platforms in industrial fields such as manufacturing, logistics, and energy management, reducing the technical threshold and trial-and-error costs for enterprises to develop their own systems. Second, the fair allocation mechanism based on Shapley values provides an objective and verifiable solution to the long-standing disputes over contribution measurement and benefit distribution in inter-enterprise model exchange and collaboration, which helps to establish a mutually trusting AI ecosystem partnership. Third, the integrated privacy protection framework of differential privacy and SMPC provides a technically feasible path for enterprises to participate in collaborative AI while meeting strict compliance requirements such as GDPR and data security

laws, bridging the gap between compliance requirements and technical implementation.

6.3 Research Limitations and Future Work

While achieving the aforementioned results, this study also has several limitations that deserve to be acknowledged, which provide direction for future research.

First, there are limitations in the scale of the experiments. Current experiments are conducted in an environment with a maximum of 100 simulated nodes. While this covers typical small- to medium-sized industrial scenarios, it is still orders of magnitude smaller than the actual deployment scale of thousands of edge devices in large multinational manufacturing enterprises. Future research needs to conduct stress tests on a real distributed testbed with 1,000+ nodes to further verify the scalability and robustness of the architecture under extreme conditions.

Second, the scope of data validation is limited. The experimental data is based on academic benchmark datasets such as MNIST and CIFAR, and validation with real-world industrial scenario data is still lacking. Different industrial sectors—such as optical inspection images in semiconductor manufacturing, multivariate time-series data in chemical processes, and RFID trajectory data in logistics and warehousing—have vastly different statistical characteristics and domain knowledge, requiring targeted validation. Future longitudinal studies should be deployed and evaluated in the real production environments of partner companies.

Third, the assumptions of economic models need to be gradually relaxed. Current mechanism designs are based on standard assumptions such as risk neutrality, single-period static contracts, and perfect rationality. In real markets, complex factors such as participants' risk aversion preferences, strategic behavior under multi-period dynamic contracts, and competition and collusion among market participants can significantly affect incentive effects. Introducing behavioral game theory and dynamic mechanism design theory is an important direction for improving the model's explanatory power in reality.

Fourth, there is a lack of adaptability. Currently, the adjustment of incentive weights relies on predefined heuristic scheduling strategies, lacking the ability to perceive and adapt to dynamic changes in the market (such as fluctuations in model supply and demand, the influx of new participants, and technological paradigm shifts). Adaptive incentive tuning mechanisms based on multi-agent reinforcement learning (MARL) hold promise for achieving better long-term social welfare in dynamic and uncertain environments, and represent a highly promising research frontier.

Fifth, the challenges of standardization and interoperability. Industry standardization of AI model metadata models, quality assessment benchmarks, and incentive settlement protocols is a prerequisite for promoting the proposed system from academic prototypes to industrial applications. This requires collaborative efforts from academia, industry, and standardization organizations (such as ISO/IEC JTC 1/SC 42 Artificial Intelligence Subcommittee).

Despite the aforementioned limitations, this study systematically constructs a theoretical framework, architectural design, and

experimental verification system for an AI model trading market oriented towards Industry 5.0, laying a solid academic foundation for the construction of a safe, fair, and sustainable industrial AI ecosystem. With the accelerated integration of edge intelligence, Web3, and generative AI technologies, the concept of AI models as tradable digital assets will become a reality in the near future, and the results of this research are expected to play a positive and far-reaching role in this historic process.

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ABSTRACT

Industry 5.0 aims for deep human-machine collaboration with human-centricity, sustainability, and resilience as core values. In this paradigm shift, AI models trained on edge devices have emerged as critical digital assets for enterprises, yet a secure, fair, and efficient framework for sharing and trading such models remains underdeveloped. This study proposes a comprehensive architecture and incentive mechanism for an AI model exchange market tailored to Industry 5.0 environments.

A five-layer hybrid architecture integrating edge computing, federated learning, blockchain, differential privacy, and secure multi-party computation is designed. A two-stage incentive mechanism combining contract theory and Shapley value-based distribution is formalized. Extensive experiments on MNIST, Fashion-MNIST, CIFAR-10, and CIFAR-100 demonstrate that the proposed method achieves up to 3.17pp higher accuracy than FedAvg, reduces the Gini coefficient to 0.285, and cuts MIA attack success rate by 82.0% with DP ($\epsilon=1.0$). The prototype sustains 820 TPS with 95.8% success rate across 100 nodes.

This work provides both theoretical foundations and practical guidelines for building secure, fair, and sustainable industrial AI ecosystems in the Industry 5.0 era.

Keywords: Industry 5.0, AI Model Exchange Market, Federated Learning, Blockchain, Incentive Mechanism, Differential Privacy, Shapley Value